



Ahmad Ghalawinji

DATA ENGINEER & AI SPECIALIST

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Data engineer and AI specialist with industry experience building ML pipelines and data systems at scale.
Seeking data engineering or applied AI roles.

Education

ENSIMAG and Grenoble Alpes University

MSC IN DATA SCIENCE AND ARTIFICIAL INTELLIGENCE (MoSIG M2)

Grenoble, France

Sep. 2023 - Sep. 2024

- Relevant coursework: Advanced Machine Learning, Natural Language Processing, Big Data Management

ENSIMAG and Grenoble Alpes University

MSC IN INFORMATICS (MoSIG M1)

Grenoble, France

Sep. 2022 - Sep. 2023

- Relevant coursework: Algorithms, Software Engineering, Operating Systems

Tishreen University

BSc IN COMPUTER ENGINEERING

Latakia, Syria

Sep. 2016 - Sep. 2021

- Relevant coursework: Data Structures, Computer Architecture, Database Systems

Experience

Self-Employed

FREELANCE WEB & INFRASTRUCTURE ENGINEER

Remote, France

Jan. 2025 - Present

- Deployed and managed Linux/WordPress infrastructure for multiple clients with high uptime achieved through proactive monitoring, SSL certificate management, and automated backups.
- Configured Nginx reverse proxies, DNS management, and Git-based deployment workflows to enable rapid iterations and streamlined development processes.
- Technologies:** Linux, Nginx, MySQL, Git, WordPress, SSL/TLS

Amadeus

ARTIFICIAL INTELLIGENCE ENGINEERING INTERN

Nice, France

Feb. 2024 - Aug. 2024

- Improved production forecasting system by designing data pipeline in Databricks to integrate external pricing signals into the machine learning model.
- Achieved 4% improvement in forecast accuracy, resulting in 0.4% revenue increase.
- Technologies:** Python, MongoDB, Azure, Spark on Databricks

Inria

MACHINE LEARNING INTERN

Grenoble, France

Feb. 2023 - Jul. 2023

- Conducted research into deep neural models including transformers and variational autoencoders (VAEs) to enhance representation learning.
- Utilized PyTorch to develop VAE that reduced data dimensionality, enabling study of dynamic system behavior.
- Technologies:** Python, PyTorch, VAEs

Laboratoire d'Informatique de Grenoble (LIG)

SOFTWARE ENGINEERING INTERN

Grenoble, France

Feb. 2022 - Jul. 2022

- Defined automated planning workflows and implemented transition from Domain-Specific Language (DSL) to automated planning using meta-model for Petri nets.
- Technologies:** DSL, Modelling

Projects

TREND ORACLE — NEWS Q&A SYSTEM WITH RAG | GitHub | Report

2026

- Built a fully local news Q&A system collecting articles from five international RSS feeds with NLP pipelines for sentiment analysis and named entity recognition.
- Designed retrieval-augmented generation (RAG) architecture using MiniLM embeddings and ChromaDB, enabling natural-language queries over collected news with cited sources.
- Engineered complete data pipeline (collection, processing, indexing) with DuckDB for structured queries and Docker Compose containerization.
- Delivered interactive Gradio interface with streaming responses, clickable source links, and live sidebar for article exploration.
- Technologies:** Python, spaCy, HuggingFace Transformers, ChromaDB, DuckDB, Ollama, Gradio, Docker

- Investigated climate warming effects on youth cardiorespiratory fitness across 28 European countries using ERA5 climate reanalysis and socioeconomic indicators.
- Trained predictive models (Random Forest, XGBoost) showing 4.7% baseline CRF decline under +3°C warming; mitigation analysis revealed 55% recovery potential through enhanced policies.
- Built interactive Streamlit dashboard for policymakers to explore climate scenarios and evaluate adaptation strategies in real-time.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Plotly, Streamlit, ERA5 Climate Data

- Built a production-style ETL pipeline to detect vessel port call events from 27,000+ AIS records across 16 ports, using geodesic geofencing (20 km threshold) and 30-minute deduplication to minimise false positives.
- Orchestrated a two-task Airflow DAG to extract CSV data into PostgreSQL and detect port entries/exits; inferred next destinations per vessel using a SQL LEAD() window function.
- Improved test coverage with pytest, asserting correct vessel IDs, port names, timestamps, and row values — not just call counts; used tmp_path fixtures for realistic CSV integration tests.
- Identified and documented anchoring false positives; proposed mitigation via speed threshold (<0.5 knots) and positional drift analysis to discriminate anchoring from true port docking.
- Delivered an interactive Streamlit dashboard with multi-dimensional filters and summary metrics; containerised the full stack with Docker Compose.
- **Technologies:** Python, PostgreSQL, Apache Airflow, Streamlit, Docker, Pandas, GeoPy

Skills

Programming	Python (Advanced), SQL, Java
Machine Learning & NLP	Supervised & Unsupervised Learning, NLP, RAG, ChromaDB, VAEs, Transformers, PyTorch, Scikit-learn, XGBoost
Data Engineering	Apache Spark, Apache Airflow, Azure Databricks, AWS (S3, EC2), ETL Pipelines, PostgreSQL, MongoDB, Snowflake, DuckDB, Terraform
Visualization	Streamlit, Plotly, Gradio
Tools & Deployment	Git, Linux, Docker, LaTeX

Languages

Arabic	Native
English	Advanced (C2)
French	Intermediate (B2)

Certificates

- Aug. 2024 **Azure Spark Databricks Essential Training**, LinkedIn Learning
- Aug. 2024 **ETL in Python and SQL**, LinkedIn Learning
- Aug. 2024 **Advanced NoSQL for Data Science**, LinkedIn Learning
- Aug. 2024 **Learning SnowflakeDB**, LinkedIn Learning
- Aug. 2024 **Infrastructure as Code with Terraform**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations: Infrastructure as Code**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations**, LinkedIn Learning
- Aug. 2024 **Agile Foundations**, LinkedIn Learning